

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 415 — Past Questions & Materials
by Renance

Course: EMT 415
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — EMT 415 Soil and Water Analysis 2024/2025.

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY
FIRST SEMESTER 2024/2025 UNDERGRADUATE EXAMINATION
COURSE: EMT 415: SOIL AND WATER ANALYSIS (2 UNITS)
TIME ALLOWED: 2 HOURS

INSTRUCTIONS: Answer four questions and at least one from each section

SECTION A

1. Explain the concept of cation-exchange capacity (CEC) and its significance in soil fertility. Discuss how changes in soil pH, organic matter content, and clay mineral composition can affect the CEC and, in turn, nutrient availability to plants
2. Discuss the role of soil pH in determining the availability of nutrients and its impact on plant growth. How does soil pH influence the solubility of heavy metals, and what are the implications for soil health in contaminated areas?

SECTION B

- 3a. Explain the importance of water sampling in environmental monitoring and describe any four (4) objectives of water sampling.
- b. Distinguish between grab sampling, composite sampling, and integrated sampling, highlighting one key advantage of each.
- c. Identify and briefly explain any four (4) sampling errors that may occur during water sampling, and recommend one way to prevent each.
- 4a. Outline the procedure for determining total coliform and fecal coliform using either the MPN or membrane filtration method.
- b. Describe how a field blank and a duplicate sample are used in quality control during water analysis.
- c. Explain the purpose of a chain of custody form and discuss how it helps maintain the integrity of water samples from field to laboratory.

SECTION C

5. You are appointed to investigate groundwater contamination in a community located near a petrochemical facility. Preliminary reports suspect high concentrations of cadmium, lead, and benzo(a)pyrene.